MILITARY STANDARD

MIL-STD-1325-182
(NAVY)

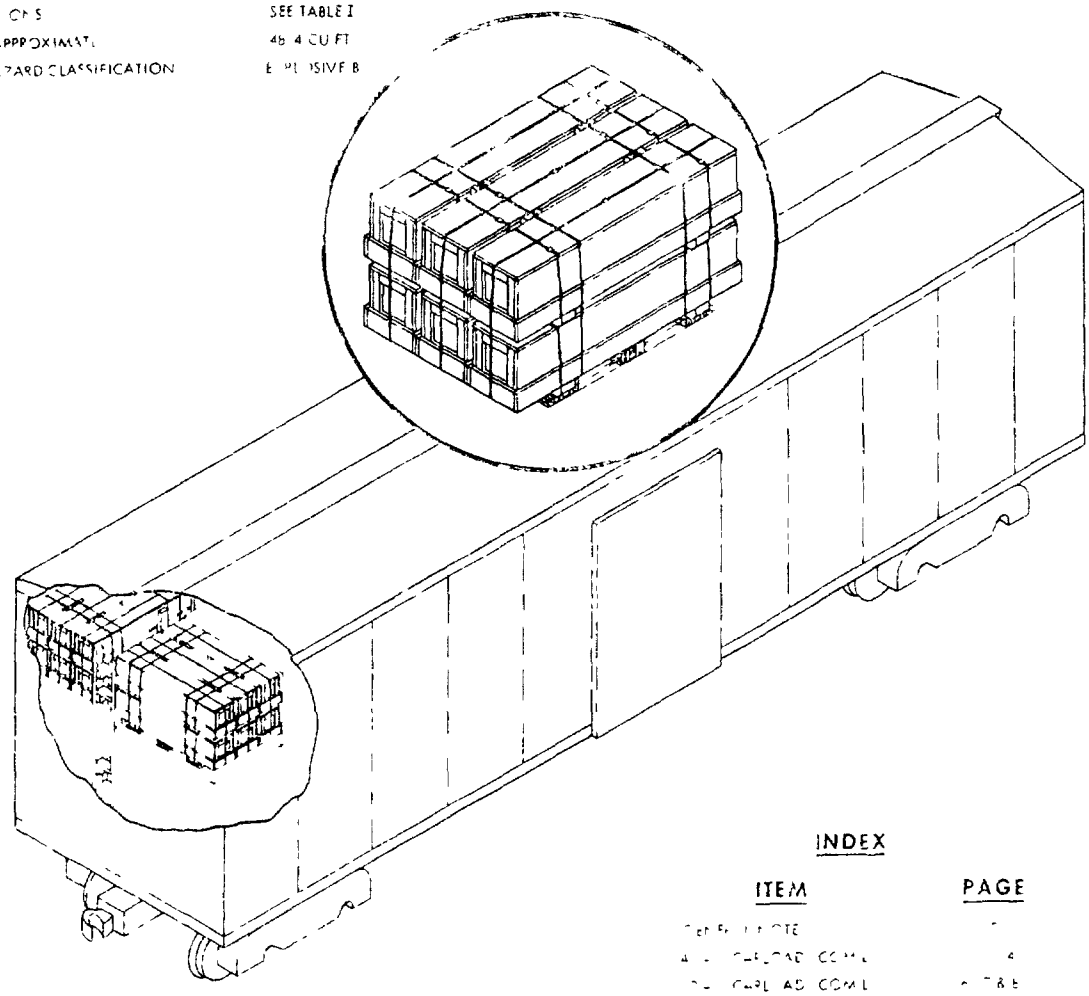
RAILCAR LOADING OF HAZARDOUS MATERIALS

12 JULY 1976

ROCKET MOTOR MK 52 AND MODS (SPARROW) OR ROCKET MOTOR MK 53 MOD 1 OR MK 78 MOD 0 (SHRIKE) FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING	WR-54 287
WEIGHT	SEE TABLE I
DIMENSIONS	SEE TABLE I
CUBE APPROXIMATE	46.4 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVE B



INDEX

ITEM	PAGE
GENERAL NOTE	2
ALL RAILROAD COMPANY	4
ALL RAILROAD COMPANY	A-1 & B

NOTE

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. SEE REFERENCED TO ASSOCIATED PALLETIZING TRUCK
3. SEE DOD CONTAINERLOADING MILITARY STANDARDS REFERENCED IN MIL-HDBK-236

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE

TECHNICAL MISSION AGENCY (NAVY) — DATE 5/30/76

COMSYSCOM BY DIRECTION — DATE

APPROVED BY BUREAU OF EXPLOSIVES

5/6/76
FOR DIRECTOR MILITARY & INTERSTATE DELIVERIES — DATE

ORIGINATOR — DATE 5/12/76

NAVAL AMMUNITION LOADING LABORATORY
[illegible], NEW JERSEY

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GENERAL NOTES

1 FOR GENERAL INFORMATION CONCERNING ORDERING INSPECTING AND PREPARING CARS AND FOR DUNNAGING MATERIALS DESIGN AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) RAILCAR LOADING OF HAZARDOUS MATERIALS

2 WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT DODX UTILITY LOADER CARS SHALL NOT BE USED FOR ITEMS WITH EXPLOSIVES A OR B HAZARD CLASSIFICATION

3 LOADING PLANS SHOWN ARE FOR COMMERCIAL BOXCARS WITH 40 FT 6 IN INSIDE LENGTH 110 IN INSIDE WIDTH AND COMMERCIAL BOXCARS WITH 50 FT 6 IN INSIDE LENGTH, 110 IN INSIDE WIDTH

4 THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS

5 IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR

6 THE LOAD CONSISTS OF ROCKET MOTORS MK 52 AND MODS (SPARROW) OR ROCKET MOTOR MK 53 MOD 1 OR MK 78 MOD 0 (SHRIKE) PALLETIZED IN ACCORDANCE WITH WR-54/287

7 THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORKLIFT TRUCK

8 UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY)

9 ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH TYPE JOINT THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971

10 APPLICABLE MATERIAL SPECIFICATIONS

DUNNAGE LUMBER - FED SPEC MM-L-751

NAILS - FED SPEC FF-N-105

STRAPPING - FED SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED)

SEALS - FED SPEC QQ-S-781, SYTLE III HEAVY DUTY

11 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL I)

40 FT 6 IN BOXCAR, COMMERCIAL

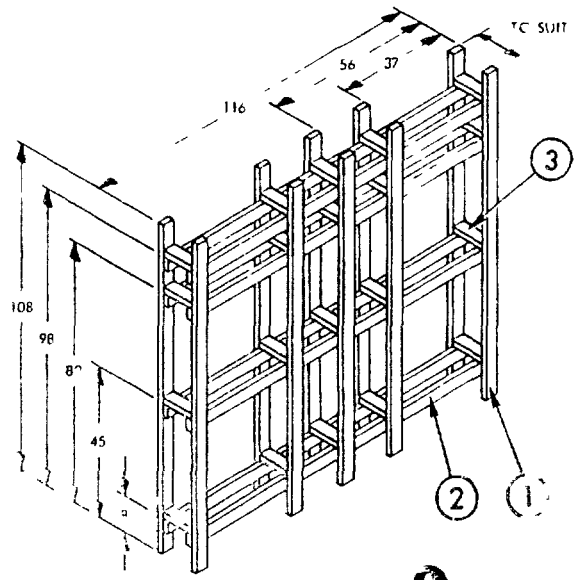
1 THE CARLOAD CONSISTS OF 57 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING

2 WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH MIL-STD-1325-102 (NAVY) SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH SEE MIL-STD-1325 (NAVY)

3 THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE

TABLE I

ROCKET MOTOR			UNIT LOAD	UNIT LOAD DIMENSIONS		
DODIC/ NALC NO	MK	ON PALLET MK & MOD	WT (APPROX) LBS	L	W	H
V 357	52	12-0	1412	58	39	37 1/8
V 357	52	12-1	1374	58	39	36 3/4
V 334	53	12-0	1424	58	39	37 1/8
V 334	53	12-1	1386	58	39	36 3/4
V 322	78	12-0	1514	58	39	37 1/8
V 322	78	12-1	1476	58	39	36 3/4



DETAIL A

SWAY BRACE ASSEMBLY

(4 REQUIRED)

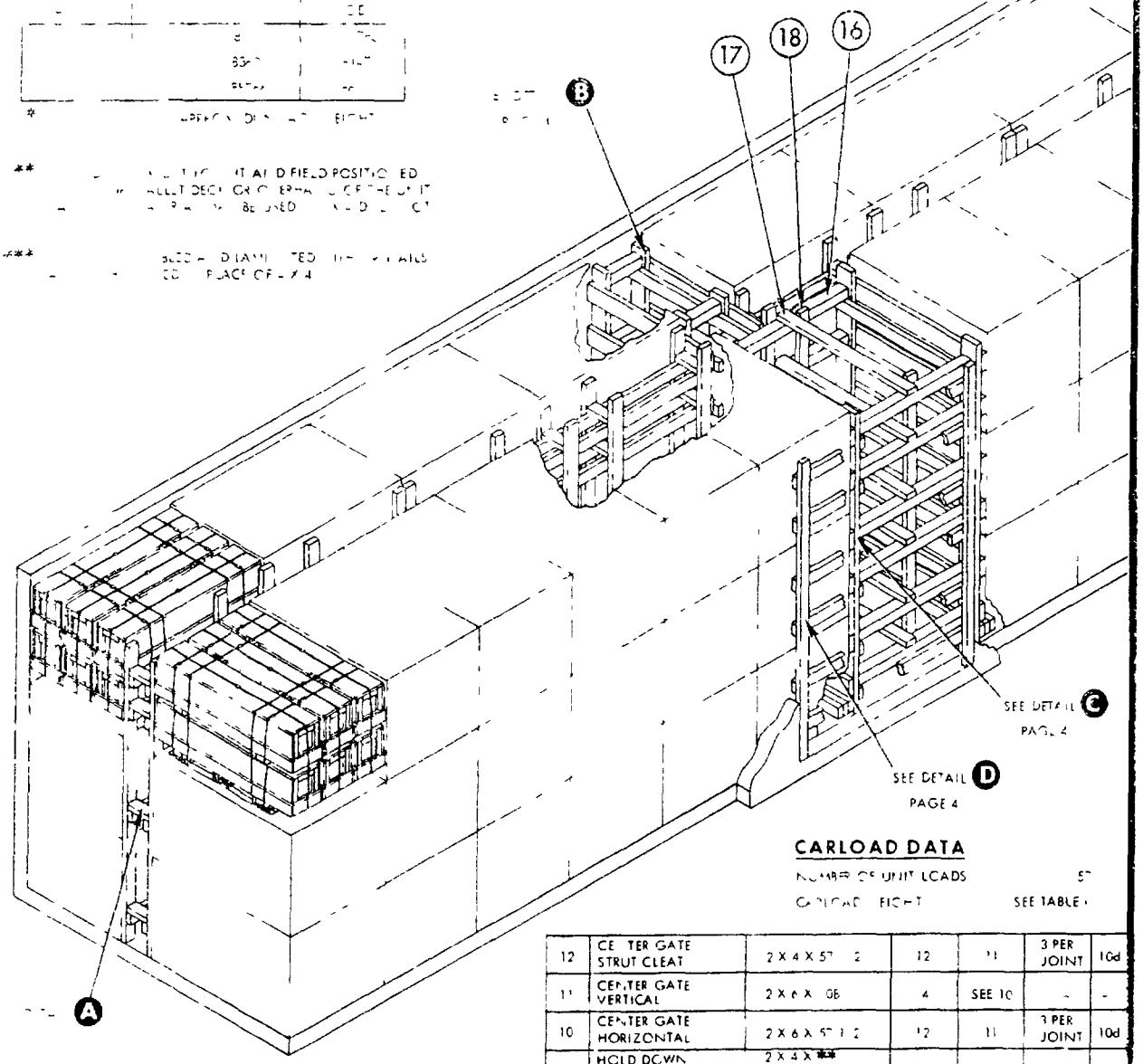
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NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
20	HORIZONTAL DOORWAY MEMBER	2 X 4 X DOOR WIDTH	12	19	3 PER JOINT	16d
19	VERTICAL DOORWAY MEMBER	2 X 3 X 108	4	DOOR POST	3 PER FOOT	20d
8	VERTICAL TIE BAR	2 X 4 X 96	2	16	2 PER JOINT	16d
1	HORIZONTAL TIE BAR	2 X 4 X 58	4	16	2 PER JOINT	16d
14	STRUT	4 X 4 X *** WEDGE FIT	-	5, 11	2 PER JOINT	16d
2	HOLD DOWN CLEAT	2 X 4 X ** TO SUIT	4	14	4	10d
4	HOLD DOWN CLEAT	2 X 4 X ** TO SUIT	4	13	4	10d
18	HOLD DOWN SPACER	2 X 4 X 57 1/2	2	11	3 PER JOINT	16d

LIST OF MATERIALS & NAILING DATA

NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
12	CENTER GATE STRUT CLEAT	2 X 4 X 57 1/2	12	11	3 PER JOINT	10d
11	CENTER GATE VERTICAL	2 X 6 X 108	4	SEE 10	-	-
10	CENTER GATE HORIZONTAL	2 X 6 X 57 1/2	12	11	3 PER JOINT	10d
9	HOLD DOWN CLEAT	2 X 4 X ** TO SUIT	2	8	4	10d
8	HOLD DOWN CLEAT	2 X 4 X ** TO SUIT	4	7	4	10d
7	HOLD DOWN SPACER	2 X 4 X CAR WIDTH-59	2	5	3 PER JOINT	16d
6	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH-59	12	5	3 PER JOINT	10d
5	CENTER GATE VERTICAL	2 X 6 X 108	4	SEE 4	-	-
4	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH-59	12	4	3 PER JOINT	10d
3	SWAY BRACE STRUT	2 X 4 X TO SUIT	80	1	3 PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 X 4 X 116	32	1	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 X 4 X 108	40	SEE 2	-	-

LIST OF MATERIALS & NAILING DATA

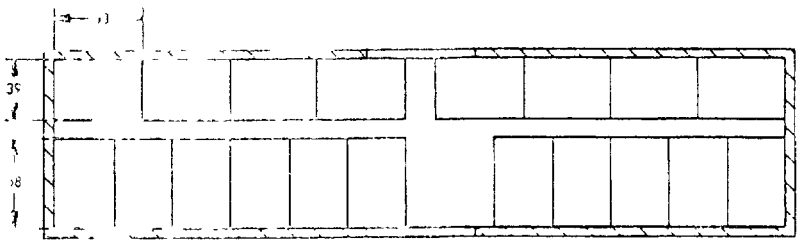


CARLOAD DATA

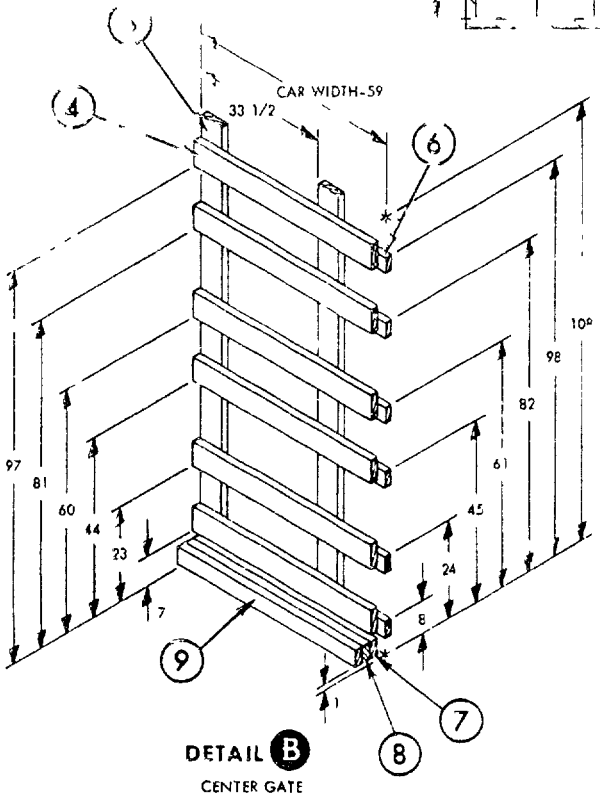
NUMBER OF UNIT LOADS 5

CARLOAD WEIGHT SEE TABLE I

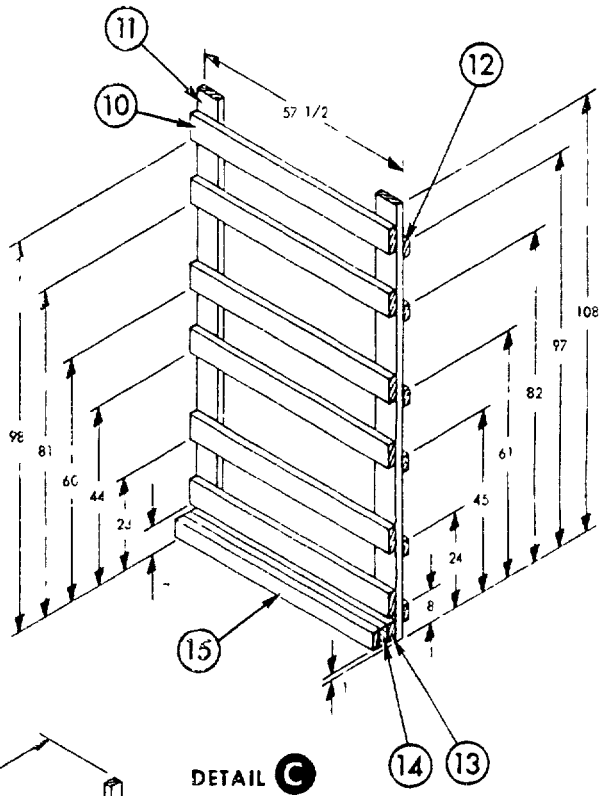
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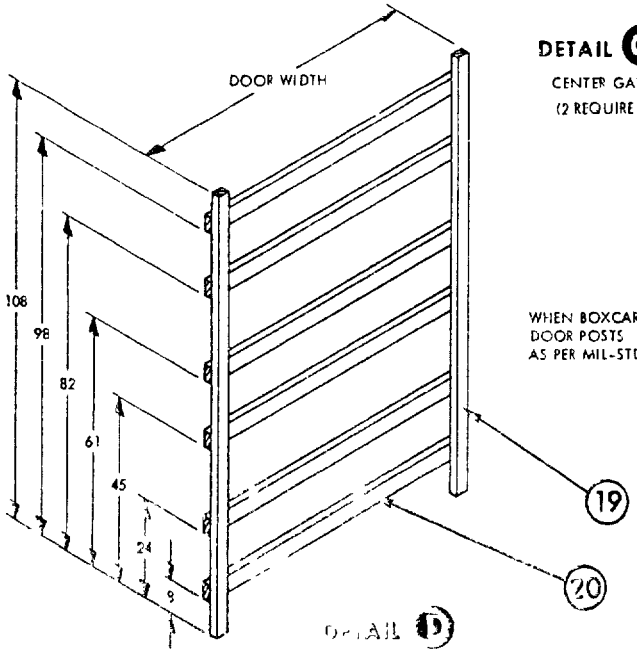
LOADING PLAN



DETAIL B
CENTER GATE
(2 REQUIRED)



DETAIL C
CENTER GATE
(2 REQUIRED)



WHEN BOXCAR HAS STEEL
DOOR POSTS DOORWAY PROTECTION
AS PER MIL-STD-1325 (NAVY) IS REQUIRED

DETAIL D
DOORWAY PROTECTION
(2 REQUIRED)

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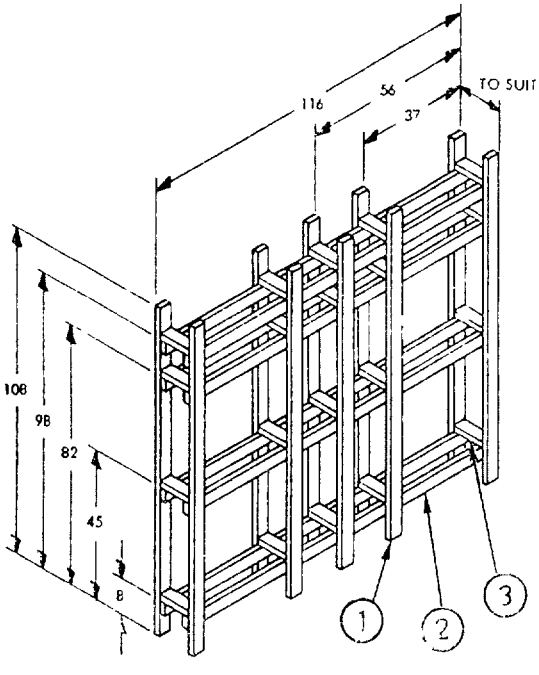
50 FT 6 IN BOXCAR, COMMERCIAL

1 THE C PLOAD CONSISTS OF 72 UNIT LOADS WHICH MUST B LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAW NG

2 WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQ RED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS THE PAR IAL LAYER OF LADING SHALL BE BRACED BY ANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN AC- ORDANCE WITH MIL-STD-1325 100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH MIL-STD-1325-102 (NAVY) SE LECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY)

3 THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CON- VENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE

4 THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG DOORS PROVIDING STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA ARE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4 STEEL STRAPS PER STACK EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE RANDOM LENGTH MATERIAL DOUBLED AND SPLICED BUT WITH JOINTS OF SPLICES OFFSET MAY BE USED DI MENSIONAL LUMBER DOORWAY PROTECTION PIECES 19 AND 20 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR THEN DIMENSIONAL LUMBER DOORWAY PROTECT- ION IS REQUIRED FOR THE CONVENTIONAL DOOR PLUG DOORS MUST BE SECURELY CLOSED AND WIRED TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER



DET A (4)

SWAY BRACE ASSEMBLY

(SHEATHING NOT SHOWN)

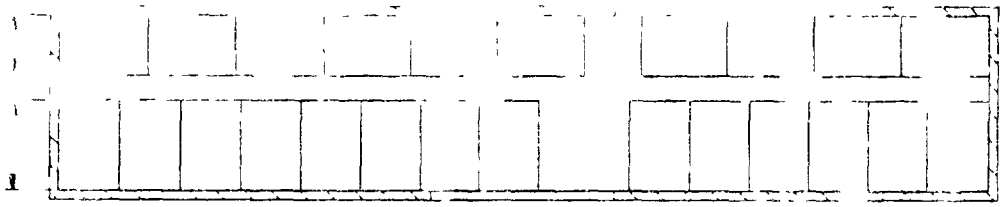
* DOUBLED 2 X 4 X CUT TO SUIT AND FIELD POSITIONED TO FIT UNDER PALLET DECK OR OVERHANG OF THE UNIT LOAD 2 X 3 MATERIAL MAY BE USED IF 2 X 4 DOES NOT FIT

** 2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4 s

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
20	HORIZONTAL DOORWAY MEMBER	2 X 4 X DOOR WIDTH	12	19	2 PER JOINT	10d
19	VERTICAL DOORWAY MEMBER	2 X 3 X 108	4	DOOR POST	2 PER FOOT	20d
18	VERTICAL TIE BAR	2 X 4 X 96	2	16	2 PER JOINT	16d
17	HORIZONTAL TIE BAR	2 X 4 X 58	6	16	2 PER JOINT	16d
16	STRUT	4 X 4 X ** WEDGE FIT	2	5 11	2 PER JOINT	16d
15	HOLD DOWN CLEAT	2 X 4 X * TO SUIT	4	14	4	10d
14	HOLD DOWN CLEAT	2 X 4 X * TO SUIT	4	13	4	10d
13	HOLD DOWN SPACER	2 X 4 X 57 1/2	2	11	3 PER JOINT	10d
12	CENTER GATE STRUT CLEAT	2 X 4 X 57 1/2	12	11	3 PER JOINT	10d
11	CENTER GATE VERTICAL	2 X 6 X 108	4	SEE 10	-	-
10	CENTER GATE HORIZONTAL	2 X 6 X 57 1/2	12	1	3 PER JOINT	10d
9	HOLD DOWN CLEAT	2 X 4 X * TO SUIT	2	8	4	10d
8	HOLD DOWN CLEAT	2 X 4 X * TO SUIT	2	7	4	10d
7	HOLD DOWN SPACER	2 X 4 X CAR WIDTH-59	2	5	3 PER JOINT	10d
6	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH-59	12	5	3 PER JOINT	10d
5	CENTER GATE VERTICAL	2 X 6 X 108	4	SEE 4	-	-
4	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH-59	12	5	3 PER JOINT	10d
3	SWAY BRACE STRUT	2 X 4 X TO SUIT	100	1	3 PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 X 4 X 116	40	1	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 X 4 X 108	50	SEE 2	-	-

LIST OF MATERIALS & NAILING DATA

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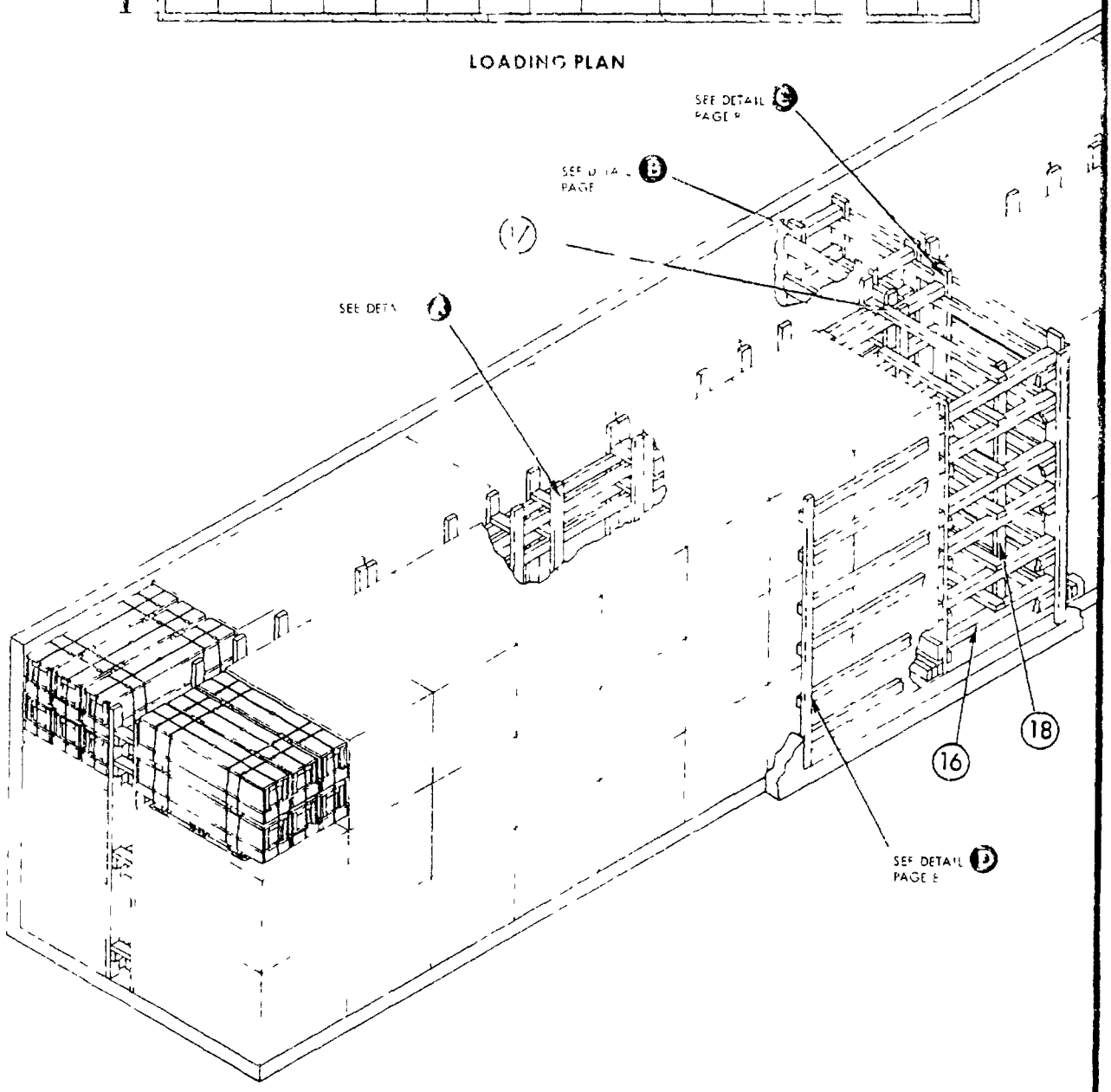


TABLE III

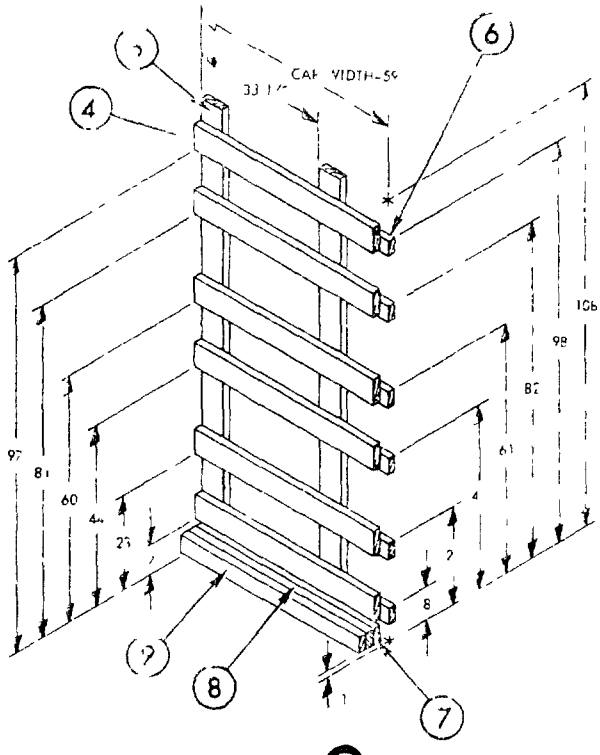
		CARLOAD WEIGHT LBS APPROXIMATE ON PALLET MIL-P-15	
	MK	OLD	METRIC
	2	104 B	101962
	53	05568	02826
	78	12042	09306

* ACTUAL DESIGN GROSS APPROXIMATE DUNNAGE WEIGHT

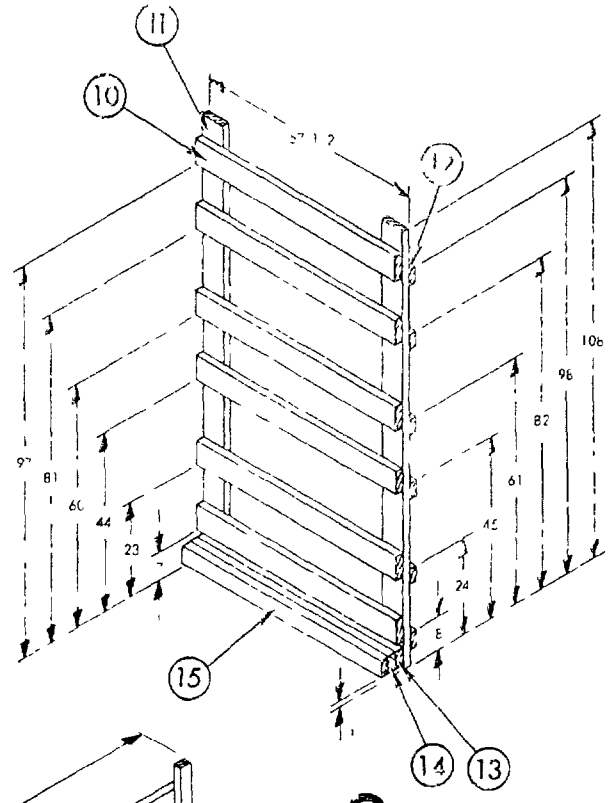
CARLOAD DATA

NUMBER OF UNIT LOADS	72
CARLOAD WEIGHT	SEE TABLE III

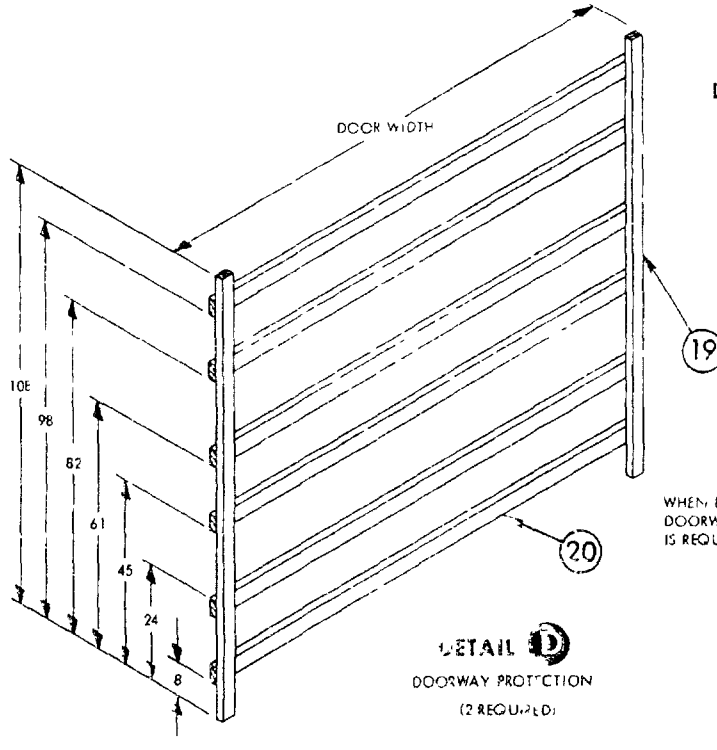
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DETAIL B
CENTER GATE
(2 REQUIRED)



DETAIL C
CENTER GATE
(2 REQUIRED)



DETAIL D
DOORWAY PROTECTION
(2 REQUIRED)

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) IS REQUIRED

REVIEW ACTIVITY
NAVY OS, AS

PREPARING ACTIVITY
NAVY OS
PROJECT NO. 3940-N[illegible]

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MTT-STD-1235-182 (NAVY)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

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